

India Petrochemicals: Carbon Targets Enter the Cracker Gate

A Bombay Breed intelligence brief for Chief Financial Officers of obligated petrochemical entities

11

petrochemical units with mandatory GEI targets

52%

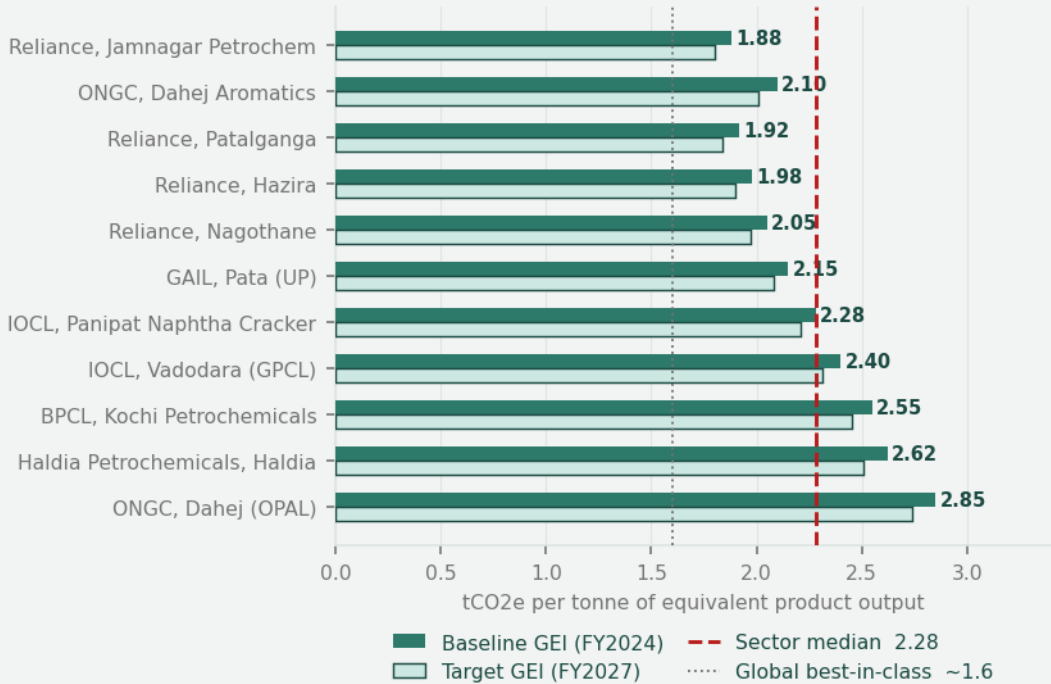
GEI spread: highest vs. lowest-intensity cracker

2x

penalty multiplier on credit shortfall

GEI LEAGUE TABLE: PETROCHEMICAL UNITS (NAPHTHA CRACKERS & AROMATICS)

Baseline (FY2023-24) vs target (FY2026-27), tCO₂e per tonne of equivalent product Source: MoEFCC GEI Target (Amendment) Rules, 13 Jan 2026

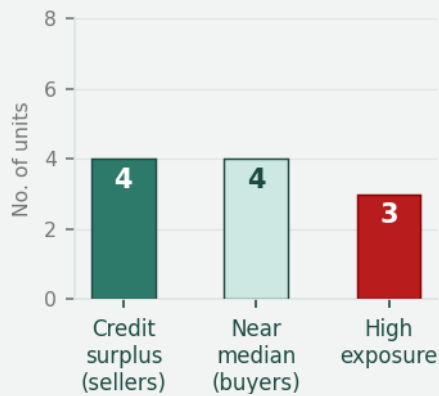


India's 11 CCTS-obligated petrochemical units span a 52% GEI intensity range, from ONGC's Dahej (OPAL) complex at **2.85 tCO₂e/tonne** — a newer but energy-intensive naphtha cracker — to Reliance Jamnagar at **1.88 tCO₂e/tonne**. Feedstock type, cracker design vintage, and integration depth drive the spread. The sector's defining dynamic: GEI targets are set at the cracker gate, not across downstream polymer lines.

Petrochemical units face a **2-3% GEI reduction** in FY2025-26, rising to **4-5%** by FY2026-27. IOCL Panipat's Naphtha Cracker alone produces 1.18 million tonnes annually — every 0.01 tCO₂e/t gap between actual GEI and target translates to **11,800 carbon credits**. At Rs 900/credit, that is Rs 1.06 crore per decimal point of performance shortfall. Reliance's multi-site portfolio creates both natural netting opportunities and consolidation risk across CCTS registration numbers.

COMPLIANCE EXPOSURE PROFILE: 11 PETROCHEMICAL UNITS

Estimated distribution. Source: MoEFCC Jan 2026; BEE CCTS Schedule



4 units

— predominantly Reliance sites with integrated feedstock access — are on track to generate surplus credits and position as market sellers.

4 units

sit near the compliance boundary and will need to optimise furnace fuel mix or purchase credits in FY2026 and FY2027.

3 units

— including older ONGC and GAIL complexes — face above-median GEI and are structurally likely credit buyers at scale.

FOUR QUESTIONS YOUR CFO SHOULD BE ABLE TO ANSWER TODAY

- 1** What is your facility-level GEI baseline under the January 2026 CCTS notification, and how does your cracker configuration rank against the 11-unit sector?
- 2** For multi-site operators (Reliance, IOCL, ONGC): what is your net credit position across CCTS registration numbers, and can surplus at one site offset deficit at another?
- 3** How does your GEI trajectory reconcile with your BRSR Scope 1 and 2 disclosures? A divergence between MoEFCC baseline and ESG-reported emissions is a board-level risk.
- 4** When the Indian Carbon Market opens mid-2026, is petrochemical credit pricing factored into your product margin models? Carbon cost is becoming a variable COGS line.

HOW BOMBAY BREED WORKS WITH OBLIGATED PETROCHEMICAL ENTITIES

- **GEI peer benchmarking:** your unit vs. all 11 obligated petrochemical entities in the CCTS schedule
- **CFO / board briefing:** CCTS carbon cost embedded in margin forecasts and product pricing models
- **Credit position modelling:** surplus or deficit in tonnes and INR, with sensitivity to feedstock GEI assumptions
- **Carbon communications:** BRSR-CCTS reconciliation and ICM launch narrative for Reliance, ONGC, IOCL investors

Request a full briefing: theresa.ronnie@bombaybreed.com · bombaybreed.com

Sources: MoEFCC Greenhouse Gases Emission Intensity Target (Amendment) Rules, 13 January 2026 | Bureau of Energy Efficiency (BEE) CCTS Compliance Schedule | IEA Chemicals 2024 | IOCL Panipat Naphtha Cracker Annual Report FY2024 | Carbon credit pricing: BEE / market data FY2026